

DATA 615 - Regression: Homework - 1

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Q.1. (5 pts)

Yes, “I have read Lab 1. I have installed relevant software, and am ready to use software for homework, labs, and data analysis.”

Q.2, Problem 1.20 (10 pts)

```
# Load data
cm <- read.table("./data/CH01PR20.txt", header = F)
colnames(cm) <- c("time", "copiers")
glimpse(cm)
```

```
## Rows: 45
## Columns: 2
## $ time    <int> 20, 60, 46, 41, 12, 137, 68, 89, 4, 32, 144, 156, 93, 36, 72, ~
## $ copiers <int> 2, 4, 3, 2, 1, 10, 5, 5, 1, 2, 9, 10, 6, 3, 4, 8, 7, 8, 10, 4, ~
```

```
summary(cm)
```

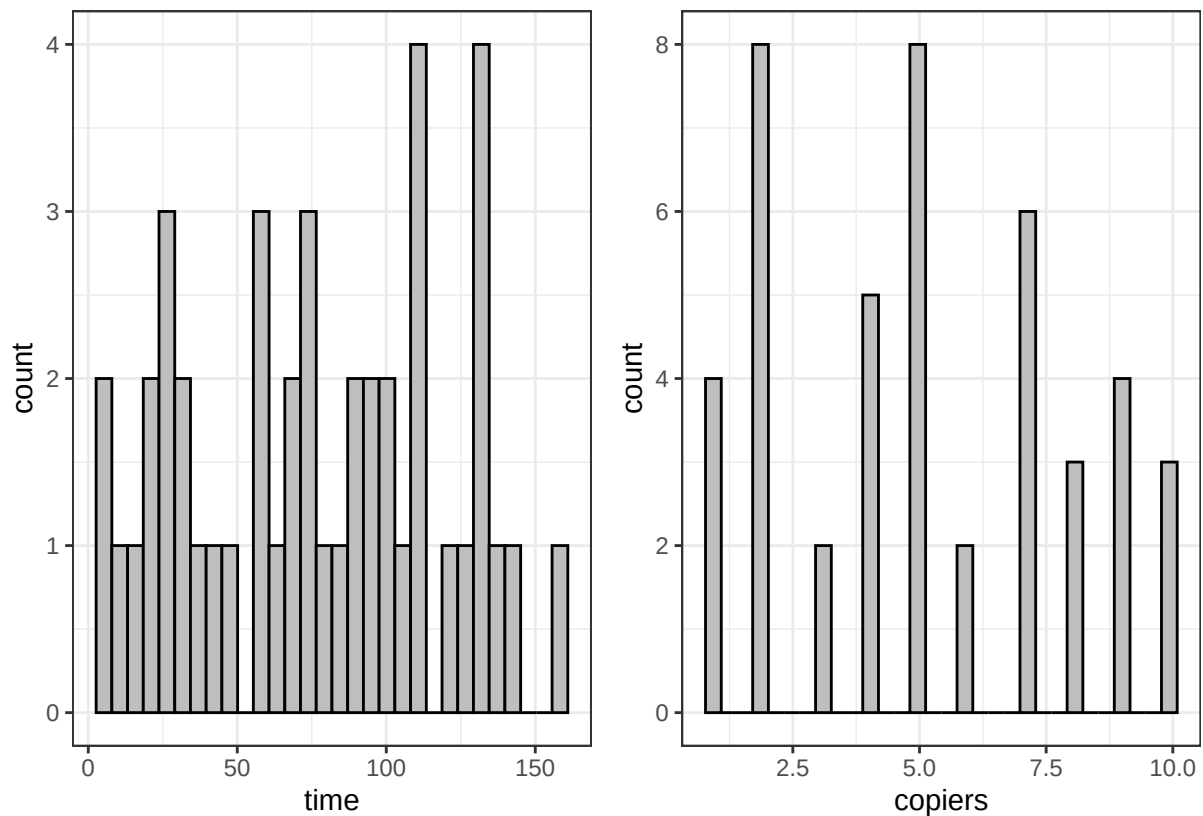
```
##           time           copiers
## Min.      : 3.00    Min.      : 1.000
## 1st Qu.: 36.00    1st Qu.: 2.000
## Median : 74.00    Median : 5.000
## Mean     : 76.27    Mean      : 5.111
## 3rd Qu.:111.00    3rd Qu.: 7.000
## Max.     :156.00    Max.       :10.000
```

Observation: For each call, X is the number of copiers serviced and Y is the total number of minutes spent by the service person.

Graphical summary (optional, not part of HW)

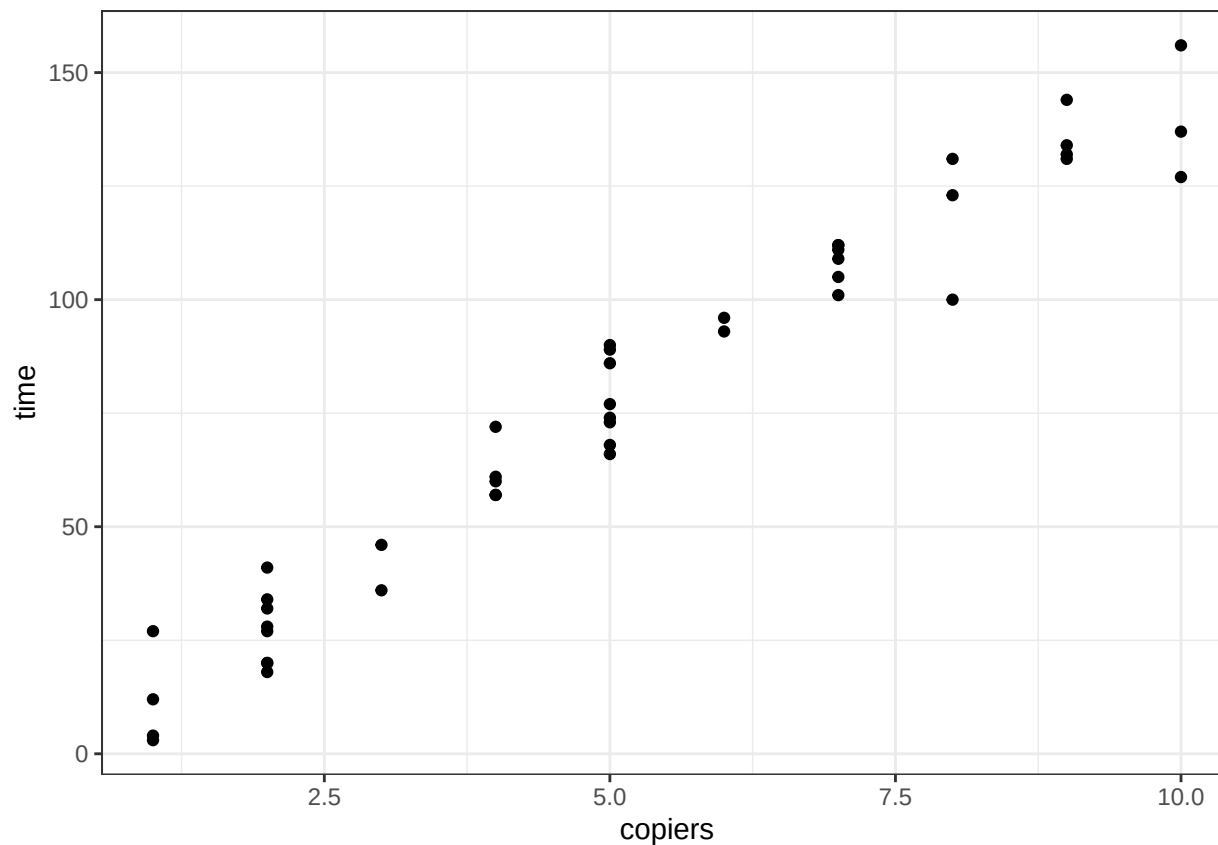
```
ggplot(cm, mapping = aes(x = time)) +  
  geom_histogram(fill = "gray", color = "black") + theme_bw() -> g1  
  
ggplot(cm, mapping = aes(x = copiers)) +  
  geom_histogram(fill = "gray", color = "black") + theme_bw() -> g2  
  
(g1 + g2)
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.  
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```



- Findings:
 - Variable `time` seems to be somewhat uniformly distributed and with no visible outliers.
 - Variable `copiers` seems to be a little skewed to the right and with no visible outliers.

```
ggplot(cm, mapping = aes(x = copiers, y = time)) +  
  geom_point() +  
  theme_bw()
```



- Findings:

- The variables `time` and `copiers` present a positive and strong linear correlation. I.e, as `copiers` increase, `time` also increases.

a. Estimated regression function (You do NOT have to copy the questions. I often include a brief description for my convenience.)

```
cm_SLR <- lm(data = cm, time ~ copiers)
summary(cm_SLR)
```

```
##
## Call:
## lm(formula = time ~ copiers, data = cm)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -22.7723  -3.7371   0.3334   6.3334  15.4039
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.5802     2.8039  -0.207   0.837
## copiers       15.0352     0.4831  31.123 <2e-16 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.914 on 43 degrees of freedom
## Multiple R-squared:  0.9575, Adjusted R-squared:  0.9565
## F-statistic: 968.7 on 1 and 43 DF,  p-value: < 2.2e-16
```

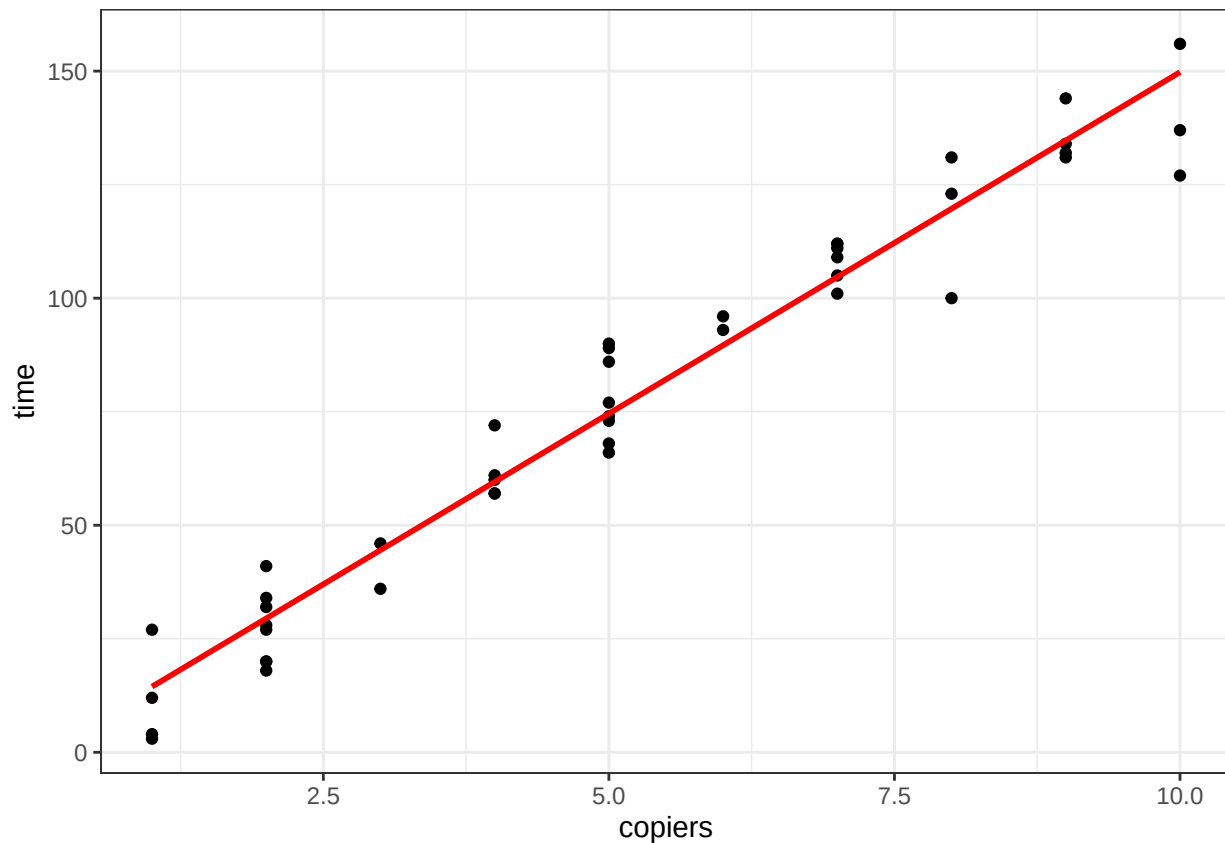
The estimated regression line (function) is:

$$\widehat{time} = -0.5802 + 15.0352(copiers)$$

b. Plot data with regression line

```
ggplot(cm, mapping = aes(x = copiers, y = time)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE, colour = "red") +
  theme_bw()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



The above plot shows indeed that the variables `copiers` and `time` hold a strong linear relationship. In fact, based on the linear model adjusted $R - squared = 0.9565$, 95% of the total variation in `time`, can be explained by the simple linear regression with `copiers` as the predictor.

c. Interpret the intercept

- The estimated intercept is $\beta_0 = -0.5802$. This means the predicted value of **time** when the value of **copiers** is equal to zero.
- The estimated slope is positive. This means that for every unit increase in **copiers**, the **time** is expected to increase 15.0352.

d. Predict y at x = 5.

$$\widehat{time} = -0.5802 + 15.0352(5) = 74.60$$

Q.3. Problem 1.24 (5 pts)

a. Residuals and Sum of the squared residuals

```
cm_SLR$residuals
```

```
##          1          2          3          4          5          6
## -9.4903394  0.4391645  1.4744125  11.5096606 -2.4550914 -12.7723238
##          7          8          9         10         11         12
## -6.5960836 14.4039164 -10.4550914  2.5096606  9.2629243  6.2276762
##         13         14         15         16         17         18
##  3.3686684 -8.5255875 12.4391645 -19.7018277  0.3334204 11.2981723
##         19         20         21         22         23         24
## -22.7723238 -2.5608355 -8.5960836 -3.6665796  4.3334204 -0.5960836
##         25         26         27         28         29         30
## -0.7370757  7.3334204 -11.4903394 -1.5960836  6.3334204  6.3686684
##         31         32         33         34         35         36
##  3.2981723 15.4039164 -9.4903394 -1.4903394 -11.4550914 -2.5608355
##         37         38         39         40         41         42
## 11.4039164 -2.7370757  7.3334204 12.5449086 -3.7370757  4.5096606
##         43         44         45
## -2.4903394  1.4391645  2.4039164
```

```
anova(cm_SLR)
```

```
## Analysis of Variance Table
##
## Response: time
##          Df Sum Sq Mean Sq F value    Pr(>F)
## copiers    1  76960   76960   968.66 < 2.2e-16 ***
## Residuals 43    3416     79
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

As provided by the ANOVA output above, the sum squared of the residuals is 3.416 with 43 degrees of freedom.

b. Estimate σ^2 and σ .

The estimated variance of the error is $MSE = 3.416/43 = 79.45$. Therefore, the estimated standard deviation of the model is $\sqrt{MSE} = \sqrt{79.45} = 8.9135$.